

CREDIT CARD FRAUDULENT DETECTION USING AN OPTIMIZED LIGHT GRADIENT BOOSTING MACHINE

MINI PROJECT REPORT

Submitted to the APJ Abdul Kalam Technological University in partial fulfillment
of requirements for the award of degree

Bachelor of Technology

in

Computer Science and Engineering (Data Science)

by

ANAND P. K. (JCE23CS006)

APARNA K. P. (JCE23CS012)

KIRAN C. (JCE23CS025)

LAVANYA M. (JCE23CS035)

Guided by

Ms. RAMYA T. V.

Assistant Professor



**DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
(DATA SCIENCE)**



JAWAHARLAL COLLEGE OF ENGINEERING AND TECHNOLOGY

LAKKIDI, MANGALAM P.O, PALAKKAD, KERALA – 679301

MARCH 2026

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JAWAHARLAL COLLEGE OF ENGINEERING AND TECHNOLOGY
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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
(DATA SCIENCE)

CERTIFICATE

This is to certify that the Mini Project report titled "**Credit Card Fraudulent Detection Using Optimized Light Gradient Boosting Machine**" submitted by **ANAND P. K.** (JCE23CS006), **APARNA K. P.** (JCE23CS012), **KIRAN C.** (JCE23CS025) and **LAVANYA M** (JCE23CS035) to the APJ Abdul Kalam Technological University, Kerala in partial fulfillment of the B.Tech. Degree in Computer Science and Engineering (Data Science) is a bonafide record of the Mini Project work carried out by him/her under the guidance **Ms. RAMYA T. V.** This report in any form has not been submitted to any other University or Institute for any purpose.

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PSO2: Interpret the basic concepts and methods of computer systems and technical specifications to provide accurate solutions.

PSO3: Apply theoretical and practical proficiency with a wide area of programming knowledge, design new ideas and innovations towards research.

DECLARATION

We undersigned hereby declare that the Mini Project report “**Credit Card Fraud Detection Using Optimized Light Boosting Machine**”, submitted for partial fulfillment of the requirements for the award of degree of Bachelor of Technology of the APJ Abdul Kalam Technological University, Kerala, is a Bonafide work done by us under the guidance of **Ms. Ramya T. V.**, Assistant Professor, Department of Computer Science and Engineering, Jawaharlal College of Engineering and Technology. This submission represents our ideas in our own words and where ideas or words of others have been included; we have adequately and accurately cited and referenced the original sources. We also declare that we have adhered to ethics of academic honesty and integrity and have not misrepresented or fabricated any data or idea or fact or source in our submission. We understand that any violation of the above will be a cause for disciplinary action by the institute and/or the University and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been obtained. This report has not been previously formed the basis for the award of any degree, diploma or similar title of any other University.

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ABSTRACT

Due to the surge of interest in online retailing, the utilization of credit cards has been quickly extended for a long time. Taking the card details to perform online exchanges, which is called extortion, has moreover seen more habitually. New advances in electronic commerce systems and communication technologies have made the credit card the potentially most popular method of payment for both regular and online purchases; thus, there is significantly increased fraud associated with such transactions. Fraudulent credit card transactions cost firms and consumers large financial losses every year, and fraudsters continuously attempt to find new technologies and methods for committing fraudulent transactions. The detection of fraudulent transactions has become a significant factor affecting the greater utilization of electronic payment. Thus, there is a need for efficient and effective approaches for detecting fraud in credit card transactions. Here, an intelligent approach for detecting fraud in credit card transactions using an optimized light gradient boosting machine (OLightGBM). In this project, a Bayesian-based hyperparameter optimization algorithm is intelligently integrated to tune the parameters of a light gradient boosting machine (LightGBM). To demonstrate the effectiveness of the proposed OLightGBM for detecting fraud in credit card transactions, experiments were performed using real-world public credit card transaction data sets consisting of fraudulent transactions and legitimate ones.

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LIST OF ABBREVIATIONS

CNN	Convolutional Neural Network
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CHAPTER 1

INTRODUCTION

The identification of people by measuring some traits of individual anatomy, physiology or other behavioral characteristics has led to a specific research area called biometric recognition. Biometric technologies provide a strong mechanism for authentication and are still under continuous development. Their diffusion is mainly supported by governments, forensics and law enforcement agencies with the aim of improving the public security or in general a sense of security; in fact, the biometric identification does not directly improve the security but acts as deterrent to illegal activities.

Fingerprint recognition has been successfully used by law enforcement agencies to identify suspects and victims for almost 100 years. Recent advances in automated fingerprint identification technology, coupled with the growing need for reliable person identification, have resulted in an increased use of fingerprints in both government and civilian applications such as border control, employment background checks, and secure facility access. Examples of large-scale fingerprint systems in the US government arena include the US-VISIT's IDENT program and the FBI's IAFIS service.

1.1 OBJECTIVE

Finger-scan technology is the most widely deployed biometric technology, with a number of different vendors offering a wide range of solutions. Among the most remarkable strengths of fingerprint recognition, we can mention the following:

- ❖ Its maturity, providing a high level of recognition accuracy.
- ❖ The growing market of low-cost small-size acquisition devices, allowing its use in a broad range of applications, e.g., electronic commerce, physical access, PC logon, etc.

- ❖ The use of easy-to-use, ergonomic devices, not requiring complex user-system interaction.

On the other hand, a number of weaknesses may influence the effectiveness of fingerprint recognition in certain cases:

- ❖ Its association with forensic or criminal applications.
- ❖ Factors such as finger injuries or manual working, can result in certain users being unable to use a fingerprint-based recognition system, either temporarily or permanently.
- ❖ Small-area sensors embedded in portable devices may result in less information available from a fingerprint and/or little overlap between different acquisitions.

Fingerprint recognition or fingerprint authentication refers to the automated method of verifying a match between two human fingerprints. Fingerprints are one of many forms of biometrics used to identify individuals and verify their identity.

Developing an automatic solution to detect altered fingerprints is the first step in defeating fingerprint alteration. Fingerprint Identification is the method of identification using the impressions made by the minute ridge formations or patterns found on the fingertips. No two persons have exactly the same arrangement of ridge patterns. The patterns of any one individual remain unchanged throughout the life. Fingerprints offer a perfect means of personal identification. Other personal characteristics may change, but fingerprints do not. Fingerprints are now processed through Integrated Automated Fingerprint Identification System.

The main objective of this paper is to build a new system that is capable of finding fingerprint alteration due to cutting, abrading and burning fingers. In this project the altered fingerprints are matched with their unaltered mates. A matcher is specialized for altered fingerprints and linking them to their unaltered mates in the database utilizing whatever information is available in the altered fingerprints. That is it is an integrated automated fingerprint identification system (IAFIS) with altered fingerprints analysis and detection. It is a complete online application by using the help of AADHAAR and connected with a biometric device and the identification is done with both minutiae matching and ridge continuity checking. The applications involve Bio-Vote which is a biometric enabled online-voting and Track-Me which is an online biometric enabled people tracking for laborer identification.

1.2 PROBLEM DEFINITION

In Existing System, since existing fingerprint quality assessment algorithms are designed to examine if an image contains sufficient information (say, minutiae) for matching, they have limited capability in determining if an image is a natural fingerprint or an altered fingerprint. Obliterated fingerprints can evade fingerprint quality control software, depending on the area of the damage. If the affected finger area is small, the existing fingerprint quality assessment software may fail to detect it as an altered fingerprint.

The existing system has no identification capability and there is no biometric input device is used. The existing altered fingerprint recognition system gives only an output that the inputted image is altered or not. The problem of fingerprint obfuscation is a serious case for the following reasons: 1) Fingerprint-based biometric systems are much more widespread for large scale identification than any other biometric modality; 2) It is relatively easy to alter one's fingerprints using chemicals and abrasives compared to, say, one's iris or face, where a more elaborate surgical procedure may be necessary; and 3) Mutilated fingerprints are being routinely encountered by law enforcement and Immigration officials in several countries, thereby underscoring the urgency of finding a solution to this problem.

Given that the altered fingerprints are likely to be encountered in large-scale national identification or border control systems, the automatic detector must satisfy the following three requirements: 1) Given the large throughput requirement of these systems, the algorithm must be extremely fast. In other words, it should not increase the computational burden of the matcher by any significant amount. State-of-the-art Automated Fingerprint Identification Systems (AFIS) can process fingerprints at the rate of about 1 million matches per second. This implies that the feature extraction and decision rule used to automatically detect altered fingerprints must be simple. 2) In operational scenarios, the number of individuals with altered fingerprints that will be encountered by AFIS will be very small. The altered fingerprint detection algorithm should operate at a very small false positive rate, say one percent or lower. Subjects that are suspected to have altered fingerprints will go through a secondary inspection stage. 3) The altered fingerprint detector should be easily integrated into any AFIS.

In the case of voting applications electronic voting machine is the latest electoral system is being used by the government. Transportation of the equipment is the major worry in the current system. Occurrence of any damage or correction in the system makes it difficult to retrieve the information. Need of taking back the existing system to the counting center is a risky and hectic work. Moreover it takes a long time for the registration and a lot of risk is there. The user can't register online and voting from remote place to his own panchayath. The voting process only done at his/her own panchayath. There is no benefit for the user other than networking. The existing System depends on manual power.

Now a days in Kerala there are about 2.5 million (25 lakhs) semi-skilled or non-skilled migrant labourers from other States of India, mostly from Rajasthan, Orissa, Bihar, Jharkhand, Chhattisgarh, Assam and other north-eastern parts of India in search of employment and higher wages. Most of them are coming here, without any identification documents. Some labourers give false proof of identity to the company personnel when they get job. There is no proper enquiry about a labourer does not take place. This helps some criminals from other states, who are evading their identity from the society. A large number of similar cases are reported in now a days. So the need of a good system for personal identification arises.

1.3 THESIS OUTLINE

Chapter 2 explains the background which includes the literature survey, current scenario and methodologies. Chapter 3 deals with the proposed approach which includes description of work done, algorithm used and some limitations. Chapter 4 describes the design and implementation details which includes both hardware and software requirements, data design, architectural design, procedural design, testing and validations, simulation and results and performance comparison. Architectural design contains data and control flow diagrams. Procedural design contains design language description, major module description and proposed method and implementation. Chapter 5 deals with the conclusions made and chapter 6 gives a scope of future work.

CHAPTER 2

BACKGROUND

Fingerprint alteration has a long history. As early as 1933, Gus Winkler, a murderer and bank robber, was found to have altered the fingerprints of his left hand except for the thumb by slashing and tearing the flesh of the fingers. Further, the pattern type of one finger was altered from double loop to left loop. In more recent cases, a man using the name Alexander Guzman, arrested by Florida officials in 1995 for possessing a false passport, was found to have obfuscated fingerprints. After a two-week search based on manually reconstructing the damaged fingerprints and searching the FBI database containing 71 million records, the reconstructed fingerprints of Alexander Guzman were linked to the fingerprints of Jose Izquierdo who was an absconding drug criminal. His fingerprint mutilation process consisted of three steps: Making a “Z” shaped cut on the fingertip, lifting and switching two triangular skin patches, and stitching them back. In September 2005, a drug dealer named Marc George was apprehended because his limping gait as a result of surgery caught the attention of border officials. It is not just the criminals who have been found to alter their fingerprints. In December 2009, a woman successfully evaded the Japanese immigration AFIS by surgically swapping fingerprints of her left and right hands. Although she was originally arrested for faking a marriage license, scars on her hands made the police suspicious.



Fig. 2.1: Inked impressions before and after fingerprint alteration of Jose Izquierdo.

Fingerprint alteration has even been performed at a much larger scale involving a group of individuals. It has been reported that hundreds of asylum seekers had cut, abraded, and burned their fingertips to prevent identification by EURODAC, a European Union-wide fingerprint system for identifying asylum seekers. Although the number of publicly disclosed cases of altered fingerprints is not very large, it is extremely difficult to estimate the actual number of individuals who have successfully evaded identification by fingerprint systems as a result of fingerprint alteration. Almost all the people identified as having altered their fingerprints were not detected by AFIS, but by some other means.

2.1 LITERATURE SURVEY

In the field of fingerprint recognition, there are several papers which give different methods for identification. Some of them are easy to implement. This paper is following some of the ideas which are taken from these research papers. These ideas are very useful in this project upto some extend. The paper named Altered fingerprints; analysis and detection [1] gives a good method to find the altered fingerprints through a detailed analysis. This paper deals with the impact of the fingerprint alteration on the matching performance. It gives an idea about the three different categories of altered fingerprints with their potential countermeasures. The widespread deployment of Automated Fingerprint Identification Systems (AFIS) in law enforcement and border control applications has heightened the need for ensuring that these systems are not compromised. While several issues related to fingerprint system security have been investigated, including the use of fake fingerprints for masquerading identity, the problem of fingerprint alteration or obfuscation has received very little attention. Fingerprint obfuscation refers to the deliberate alteration of the fingerprint pattern by an individual for the purpose of masking his identity. Several cases of fingerprint obfuscation have been reported in the press. Fingerprint image quality assessment software (e.g. NFIQ) cannot always detect altered fingerprints since the implicit image quality due to alteration may not change significantly. The main contributions of this paper are: 1) Compiling case studies of incidents where individuals were found to have altered their fingerprints for circumventing AFIS, 2) Investigating the impact of fingerprint alteration on the accuracy of a commercial fingerprint matcher, 3) Classifying the alterations into three major categories and suggesting

possible countermeasures, 4) Developing a technique to automatically detect altered fingerprints based on analyzing orientation field and minutiae distribution, and 5) Evaluating the proposed technique and the NFIQ algorithm on a large database of altered fingerprints provided by a law enforcement agency. Experimental results show the feasibility of the proposed approach in detecting altered fingerprints and highlight the need to further pursue this problem.

The research paper On matching altered fingerprints [5] was supported by a grant from the FBI's Biometric Center of Excellence. In this paper they investigate the capability of a state-of-the-art commercial fingerprint matcher to match altered fingerprints to their pre-altered mates by removing spurious minutiae in the altered region. They also attempt to restore minutiae structure in a well-known type of fingerprint alteration, called 'Z'-cut, which tears local skin patches of a finger and switches them with each other. Non rigid skin distortion introduced during switching of local patches is modeled by thin-plate spline. Experimental results show that removing spurious minutiae in the altered region and relocating the minutiae in the restored local patches improve the matching performance and help to retrieve correct mates from a large background fingerprint database. In this paper, there shows that some of the altered fingerprints still have sufficient ridge information to be identified even though the severity of the alteration performed to the fingerprint appears to be large. Given a reference fingerprint database consisting of 27,000 images from NIST SD14 and 382 images from pre-altered mated fingerprint set, 86% of 1,332 altered fingerprints find their true mates within rank 100 by simply removing spurious minutiae in the invalid fingerprint region, while the minutiae templates including all the minutiae extracted from the altered fingerprint achieve 77% identification rate at rank 100. In addition, restoration of 'Z'-cut fingerprints can boost the matching performance significantly by fusing the minutiae from the altered fingerprints and their restored versions at rank level.

The paper Fingerprint matching [4] describes one of the best method of fingerprint identification or matching. Fingerprint matching has been successfully used by law enforcement for more than a century. The authors describe an automated fingerprint recognition system and identify key challenges and research opportunities in the field. This paper gives an idea about different types of sensing methods and describes a typical type of feature extraction algorithm which helps to extract minutiae templates. This paper describes a

typical type of minutiae matching algorithm also. The algorithm first uses local minutiae descriptors to coarsely align two fingerprints and then computes a global match score based on minutiae correspondences.

In the research paper named Automated Fingerprint Recognition Using the DECOC Classifier [7], describes a new method for reliable and fast feature extraction based on the classifier DECOC from skeleton fingerprint images. To identify the different shapes and types of minutiae, a Data-driven Error Correcting Output Coding (DECOC) has been adopted to work as a classifier. The proposed one has been applied throughout the fingerprint skeleton to locate various minutiae. Extracted minutiae have been used then as identification marks for an automatic fingerprint matching that is based on distance and direction between two minutiae and type of minutiae. This method classifies a block of 5x5 pixels into bifurcation and termination. The experimental results show that this method provides an acceptable true minutiae ratio, the best exchanged minutiae ratio and a good average classification time. The optimized matching method shows a good result for fingerprint recognition. The aim of this method is to identify the fingerprint without intrinsic coordinates. So, the matching distance solves the problem of rotation, displacement and the core region identification. The identification takes about 5 seconds on a 2 GB RAM, 1.66 GHz Intel Core 2 Duo processor with Windows XP operating system. This speed may be improved by hardware implementation.

In another research paper named Detection of Fingerprint Alteration Using Scars, Minutiae Density and Ridge Discontinuity [6] gives a novel method for detecting altered fingerprints using scars, minutiae density and ridge discontinuity. This paper mainly focuses on FFT enhancement for ridge discontinuity analysis and use of scars along with ridge discontinuity and minutiae density for alteration detection. The scar is detected by adaptive average filtering and thresholding. This method is found suitable for the detection of all types of alteration viz. obliteration, distortion and imitation. Parzen window method is used for constructing minutiae density map. Detection of imitation type needs further verification on real altered fingerprints. Normal fingerprints sometimes have the natural scar. Thus there is a need of method for separating scar of altered fingerprint from that of natural fingerprints.

A Comparative Study of Fingerprint Image-Quality Estimation Methods [10] gives a comparative study report on different methods of fingerprint image quality estimation. This paper reviews existing approaches for fingerprint image-quality estimation, including visual

examples showing the behavior under different quality conditions. Existing approaches have been divided into: 1) those that use local features of the image; 2) those that use global features of the image; and 3) those that address the problem of quality assessment as a classification problem.

Local and global image features are extracted using different sources: direction field, Gabor filter responses, power spectrum, and pixel intensity values. This paper also compares the behavior of the selected quality methods by reporting the correlation between them and the relationship between quality and similarity scores. For the experiments, here used the BioSec multimodal baseline corpus, which includes 19200 fingerprint images from 200 individuals acquired in two acquisition sessions using a capacitive, an optical, and a thermal fingerprint sensor. Experimental results show high correlation between genuine scores and quality, whereas almost no correlation is found between impostor scores and the quality measures. As a result, the highest improvement when rejecting low-quality samples is obtained for the false rejection rate at a given false acceptance rate.

2.2 EXISTING SYSTEM

The Automated Fingerprint Identification System (AFIS) is a biometric identification (ID) methodology that uses digital imaging technology to obtain, store, and analyze fingerprint data. The AFIS was originally used by the U.S. Federal Bureau of Investigation (FBI) in criminal cases. Lately, it has gained favor for general identification and fraud prevention.

Fingerprinting, as a form of personal identification, is a refined methodology that is proven in practice and accepted in courts of law. AFIS itself has been around for more than 25 years. Recently, a more advanced form of AFIS uses a process called plain-impression live scanning. Several vendors offer AFIS equipment and programs.

The Integrated Automated Fingerprint Identification System (IAFIS), is a national fingerprint and criminal history system in US that responds to requests to help their local, state, and federal partners and their own investigators to solve and prevent crime and catch criminals and terrorists. IAFIS provides automated fingerprint search capabilities, latent search capability, electronic image storage, and electronic exchange of fingerprints and responses. The system includes criminal histories, civil fingerprints, mostly of individuals who have

served or are serving in the U.S. military or have been or are employed by the federal government.

The main contributions of current Altered Fingerprint System are: 1) compiling case studies of incidents where individuals were found to have altered their fingerprints for circumventing AFIS, 2) investigating the impact of fingerprint alteration on the accuracy of a commercial fingerprint matcher, 3) classifying the alterations into three major categories and suggesting possible countermeasures, 4) developing a technique to automatically detect altered fingerprints based on analyzing orientation field and minutiae distribution, and 5) evaluating the proposed technique and the NFIQ algorithm on a large database of altered fingerprints provided by a law enforcement agency. Experimental results show the feasibility of the proposed approach in detecting altered fingerprints and highlight the need to further pursue this problem. The system introduced the problem of fingerprint alteration and conducted a quantitative analysis of the threat of altered fingerprints to a commercial fingerprint matcher. The system also evaluated the capability of a well-known fingerprint image quality assessment software, NFIQ, for detecting altered fingerprints. Since the NFIQ has limited ability in distinguishing altered fingerprints from natural fingerprints, The developed algorithm automatically detect altered fingerprints based on the characteristics of the fingerprint orientation field and minutiae distribution. The algorithm based on the features extracted from the orientation field and minutiae satisfies the three essential requirements for alteration detection algorithm: 1) fast operational time, 2) high true positive rate at low false positive rate, and 3) ease of integration into AFIS. The algorithm and the NFIQ criterion were tested on a large public domain fingerprint database (NIST SD14) as natural fingerprints and an altered fingerprint database provided by a law enforcement agency. At a false positive rate of 0.3 percent, the proposed algorithm can correctly detect 66.4 percent of the subjects with altered fingerprints, while 26.5 percent of such subjects are detected by the NFIQ algorithm.

CHAPTER 3

PROPOSED APPROACH

Unique i is an integrated automated altered fingerprint identification system. That is, the product is an IAFIS with altered fingerprints analysis and detection capability. Unique i is capable of finding fingerprint alteration due to cutting, abrading and burning fingers. In this project the altered fingerprints are matched with their unaltered mates. A matcher is specialized for altered fingerprints and linking them to their unaltered mates in the database utilizing whatever information is available in the altered fingerprints. Unique i is a complete online application using the concept of AADHAAR. The applications include track-me and bio-vote. Track-me is labourer identification system and bio-vote is a biometric enabled online voting system.

3.1 SYSTEM ARCHITECTURE

Unique i is an integrated automated altered fingerprint identification system. That is, the proposed system is an IAFIS with altered fingerprints analysis and detection capability. Unique i is capable of finding fingerprint alteration due to cutting, abrading and burning fingers. In this paper the altered fingerprints are matched with their unaltered mates. A matcher is specialized for altered fingerprints and linking them to their unaltered mates in the database utilizing whatever information is available in the altered fingerprints.

Unique i is a complete online application and can be linked with AADHAAR. The applications are connected with a biometric device and the identification is done with both minutiae matching and ridge continuity checking. Figure 3 shows the block diagram of Unique i. It contains 2 phases: enrollment phase and identification phase.

During the enrollment phase, the sensor scans the user's fingerprint and converts it into a digital image. The feature extractor processes the fingerprint image to identify specific details known as minutia points that are used to distinguish different users. Minutia points represent locations where friction ridges end abruptly or where a ridge branches into two or more ridges.

A typical good-quality fingerprint image contains about 20-70 minutiae points. The actual number depends on the size of the sensor surface and how the user places his or her finger on the sensor. The system stores the minutiae information—location and direction—along with the user's demographic information as a template in the enrollment database.

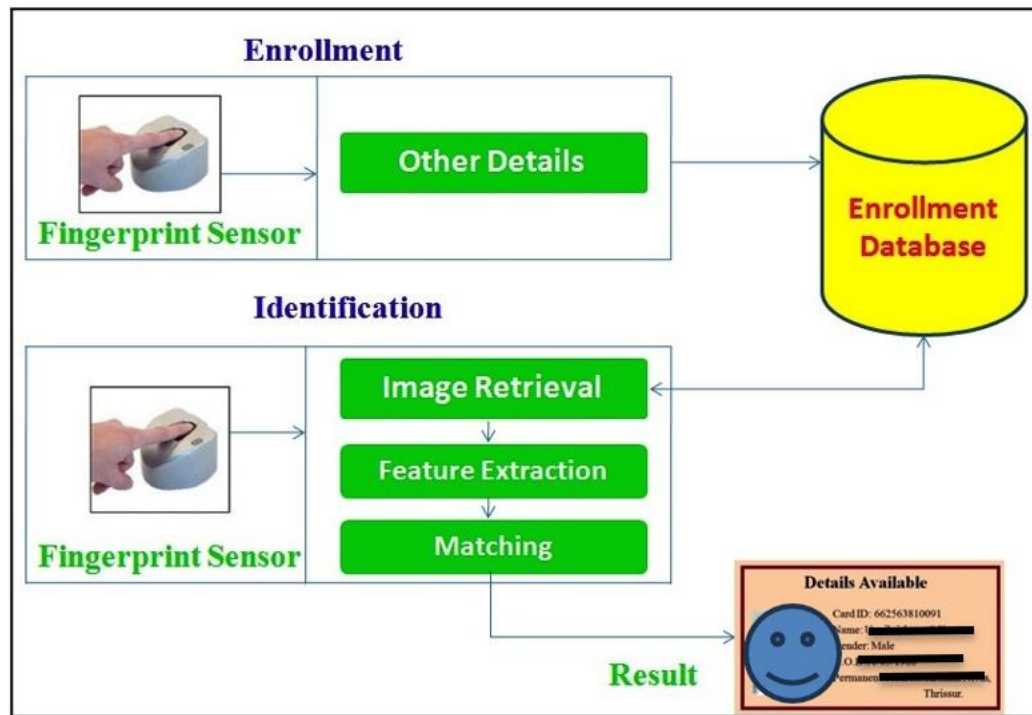


Fig. 3.1 Block-diagram of Unique i

During the identification phase, the user touches the same sensor, generating a new fingerprint image called a query print. Minutia points are extracted from the query print, and the matcher module compares the query minutia set with the stored minutia templates in the enrollment database to find the number of common minutia points. Due to variations in finger placement and pressure applied on the sensor, the minutia points extracted from the template and query fingerprints must be aligned, or registered, before matching.

After aligning the fingerprints, the matcher determines the number of pairs of matching minutiae, i.e., two minutia points that have similar location and directions. The system determines the user's identity by comparing the match score to a threshold set by the administrator.

Unique i involves 2 applications, namely Bio-Vote and Track-Me. Bio-Vote is a biometric enabled online voting application. This application can be implemented in parliament,

legislative assembly or Panchayath election. It consists of 3 levels of users, and they are Administrator, Supervisor and Voter. Administrator can create booths & candidates, register & assign supervisor, count votes etc. Supervisor can check members, view booth details etc. Voter can do vote, view candidates, results etc. Track-Me is an online biometric enabled system for laborer identification. After registering with this application a company can take details of a new labourer when he/she joins, by fingerprint identification from the Aadhaar repository. If the labourer has got a criminal history, the system will inform to the nearest police station by sending a SMS. Child labours can also find by this application.

3.2 ALGORITHM

The proposed algorithm works with the following steps;

step1. The input fingerprint sensor scans the user's fingerprint and converts it into a digital image.

Step2. Store the image and the phase = Identification, continue steps; otherwise go to step 9.

Step3. Do the feature extraction and templates are stored.

Step3.1. Input image is normalized to 512 X 480 pixels by cropping a rectangular region of the fingerprint.

Step3.2. The orientation field of the fingerprint, $\Theta(x,y)$, is computed.

Step3.3. The orientation field $\Theta(x,y)$ is approximated by a polynomial model to obtain $\hat{\Theta}(x,y)$.

Step3.4. Extract the minutiae points & minutiae density map $M_d(X)$ is computed.

step4. Compute the pair wise similarity between minutiae of the input image and other images in the database by comparing minutiae descriptors that are invariant to rotation and translation.

step5. Align two fingerprint images according to the most similar minutiae pair.

step6. Establish the minutiae correspondence.

step7. Compute a similarity score to reflect the degree of match between two fingerprints based on factors such as the number of matching minutiae, the percentage of matching minutiae in the overlapping area of two fingerprints, and the consistency of ridge count between matching minutiae.

step8. Retrieve all the information from main database and activate the application.

step9. Stop.

3.3 LIMITATIONS

Some of the limitations of Unique i is that it is suitable for only the natural and obliterated fingers. And for linking with Aadhaar needs govt. sanction.

CHAPTER 4

DESIGN AND IMPLEMENTATION

The most creative and overwhelming part of system development life cycle is system design. The design phase focuses on the detailed implementation of the system recommended in the feasibility study. System design is a transition from user-oriented document to a document-oriented to a programmer or database personal. It includes logical flow of the system. The physical design draws out of the details of the physical plan and specifies to learn the software requirements. A well-designed system should provide for controls to eliminate errors and ensure systems integrity. The design activity is often divided in two; system design and detailed design. System design aims to identify the module and how they interact with each other to produce desired results.

4.1 HARDWARE AND SOFTWARE REQUIREMENTS

- **Hardware Specification**

This refers to different hardware components used by the application.

No.	Hardware	Specification
1	PROCESSOR	PENTIUM IV
2	RAM	256 MB
3	HDD	2 GB
4	MONITOR	SVGA COLOR
5	KEY BOARD	NORMAL
6	MOUSE	NORMAL
7	FINGERPRINT READER	USB FINGY HAMSTER

Table 4.1 Hardware Specifications.

- **Software Specifications**

This refers to the different software components used for this software. Software selection is important work project development cycle. The minimum specification as follows:

No.	Type	Specification
1	OPERATING SYSTEM	WINDOWS 2000 SERVER
2	FRONT END	ASP .NET (scripting in C#)
3	BACKEND	SQL SERVER 2008
4	BROWSER	INTERNET EXPLORER 6.0
5	FINGERPRINT RECOGNITION SOFTWARE	BIOENABLE BIODESK

Table 4.2 Software Specifications.

4.2 SYSTEM ANALYSIS AND DESIGN MODELING

System Analysis and Design (SAD) Modeling is a structured process of examining a business situation (analysis), defining requirements, and creating blueprints (modeling/design) for a new or improved information system. It uses graphical, technical, and conceptual models—like UML or Data Flow Diagrams—to visualize, analyze, and document system components before coding.

4.2.1 DATA FLOW CASE DIAGRAM

4.2.2 USE CASE DIAGRAM

4.2.3 SEQUENCE DIAGRAM

4.3 MODULE DESCRIPTION

Unique i is divided into 2 main modules, and they are enrollment module and identification module. These main modules are divided into several sub Modules. The sub divisions include Image Acquisition Module, Feature Extraction Module, Matching Module and Application Module.

Enrollment Module

In enrollment module, the fingerprints are acquired using a fingerprint sensor. This is done in a sub module called image acquisition module. The captured fingerprint images are stored to a database along with the personal details such as name, gender, date of birth, age, phone number, email id, address, photo, father/husband's name, house number, place, state, district etc.

Identification Module

The identification module is subdivided into image acquisition module, feature extraction module, matching module and application module. A person is verified and identified in this module.

Image Acquisition Module

The acquisition of a fingerprint can be done off-line or on-line. In the off-line acquisition the image is typically obtained by smearing ink on the fingertip and creating an inked impression of the fingertip on the paper. After this procedure, the fingerprint is digitized by an optical scanner or a high resolution camera. This kind of fingerprint is often called rolled fingerprint. A very important kind of off-line fingerprint image is the latent fingerprint: a partial fingerprint image lifted from a crime scene by a forensic expert. Compared to a rolled fingerprint, the latent is most of the times of bad quality and hard to process. In this project we are using on-line acquisition. In the on-line acquisition, the fingerprint is acquired by using a fingerprint scanner without any kind of ink.

Feature Extraction module

Features extracted from a fingerprint image are generally categorized into three levels, as shown in Figure 4.5 Level 1 features capture macro-details such as friction ridge

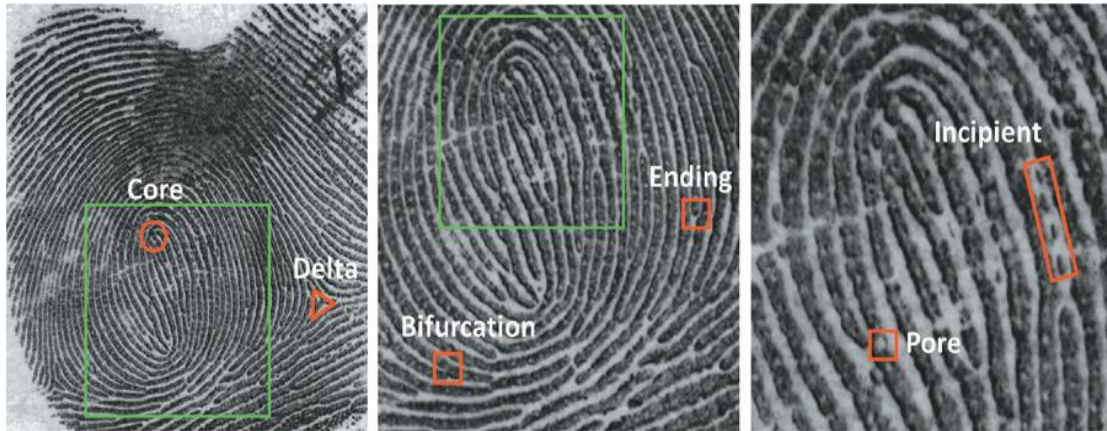


Fig:4.5 Feature levels in a fingerprint

flow, pattern type, and singular points. Level 2 features refer to minutiae such as ridge bifurcations and endings. Level 3 features include all dimensional attributes of the ridge such as ridge path deviation, width, shape, pores, edge contour, and other details, including incipient ridges, creases, and scars. Level 1 features can be used to categorize fingerprints into major pattern types such as arch, loop, or whorl; level 2 and level 3 features can be used to establish a fingerprint's individuality or uniqueness. Higher-level features can usually be extracted only if the fingerprint image resolution is high. For example, level 3 feature extraction requires images with more than 500-ppi resolution.

Minutiae Extraction

In minutiae extraction process, a series of steps are done and they are filtering, binarization and skeletonization.

- **Filtering**

In image processing filters are mainly used to suppress either the high frequencies in the image, *i.e.* smoothing the image, or the low frequencies, *i.e.* enhancing or detecting edges in the image. An image can be filtered either in the frequency or in the spatial domain. The first involves transforming the image into the frequency domain, multiplying it with the frequency filter function and re-transforming the result into the spatial domain. The filter function is shaped so as to attenuate some

frequencies and enhance others. The corresponding process in the spatial domain is to convolve the input image $f(x,y)$ with the filter function $h(x,y)$. This can be written as

$$g(x,y) = h(x,y) \odot f(x,y) \dots\dots\dots (4.1)$$

The mathematical operation is identical to the multiplication in the frequency space, but the results of the digital implementations vary, since we have to approximate the filter function with a discrete and finite kernel. The values of a given pixel in the output image are calculated by multiplying each kernel value by the corresponding input image pixel values. The output pixel value is the weighted sum of the input pixels within the window where the weights are the values of the filter assigned to every pixel of the window itself. The window with its weights is called the convolution kernel. In this project we are using convolution filtering. If the size of the kernel is $(2w+1) \times (2w+1)$ and the filter function is $h(i,j)$, then the output image $g(x,y)$ is

$$g(x,y) = \sum_{i=-w}^w \sum_{j=-w}^w h(i,j) f(x-i, y-j) \dots\dots\dots (4.2)$$

- **Binarisation**

This method consists in scanning the finger image by 10 x 10 pixels block, so each block has its threshold.

$$Threshold = \frac{\sum M(i,j)}{nl*nc} \dots\dots\dots (4.3)$$

Where $M(i,j)$ is the image matrix and $nl*nc$ is the number of pixels.

- **Skeletonisation**

Here a 3 x 3 blocks neighborhood method is used . It takes too much time for the total of the image. When the 3 x 3 block contains more than two black pixels, we apply thinning to reduce the time. In 3 x 3 window, if the central pixel is 0 and has exactly 3 zero-value neighbors, then the central pixel is a ridge branch or bifurcation. If it has only one zero-value neighbor, then the central pixel is a ridge ending or termination.

Matching Module

A fingerprint matching module computes a match score between two fingerprints, which should be high for fingerprints from the same finger and low for those from different

fingers. Fingerprint matching is a difficult pattern-recognition problem due to large intra-class variations (variations in fingerprint images of the same finger) and large interclass similarity (similarity between fingerprint images from different fingers). Intra-class variations are caused by finger pressure and placement (rotation, translation), and contact area with respect to the sensor and condition of the finger such as skin dryness and cuts. Meanwhile, interclass similarity can be large because there are only three types of major fingerprint patterns (arch, loop, and whorl).

Most fingerprint-matching algorithms adopt one of four approaches: image correlation, phase matching, skeleton matching, and minutiae matching. Minutiae-based representation is commonly used, primarily because:

- forensic examiners have successfully relied on minutiae to match fingerprints for more than a century,
- minutiae-based representation is storage efficient, and expert testimony about suspect identity based on mated minutiae is admissible in courts of law.

In this project there use both minutiae matching and continuity checking. The current trend in minutiae matching is to use local minutiae structures to quickly find a coarse alignment between two fingerprints and then consolidate the local matching results at a global level. This kind of matching algorithm typically consists of four steps, as Figure 4.6 shows. First, the algorithm computes pair-wise similarity between minutiae of two fingerprints by comparing minutiae descriptors that are invariant to rotation and translation.

Next, it aligns two fingerprints according to the most similar minutiae pair. The algorithm then establishes minutiae correspondence—minutiae that are close enough both in location and direction are deemed to be corresponding (mated) minutiae. Finally, the algorithm computes a similarity score to reflect the degree of match between two fingerprints based on factors such as the number of matching minutiae, the percentage of matching minutiae in the overlapping area of two fingerprints, and the consistency of ridge count between matching minutiae.

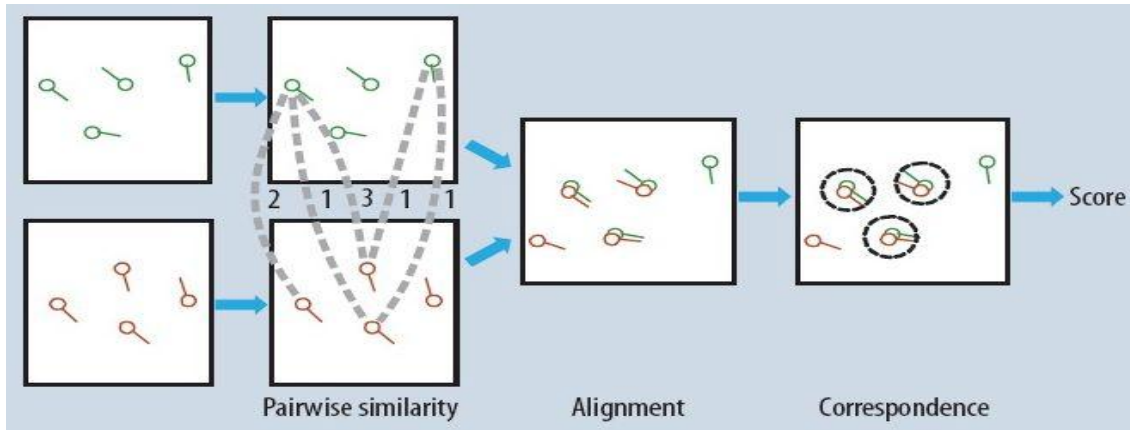


Fig:4.6 Typical minutiae-matching algorithm.

Non-Minutiae feature-based matching do not use the minutiae to perform matching but use some other features of the fingerprint ridge pattern (e.g. local ridge orientation and frequency, texture, shape). That is here we are using a continuity matching of ridge patterns. We are applying a sample signal at the beginning of a ridge and the signal passed through the ridge and received at the end. If the ridge flow breaks somewhere then we did not receive signal at the end. Actually we are not passing a signal, but checking the ridge flow by using some distance measures. One of the observed characteristics of altered fingerprints is that their ridge flow can be discontinuous in nonsingular regions as well, such as severely scarred areas, mutilated areas, and distorted ridge areas. The difference between the observed orientation field and the modeled orientation field indicates the locations and the amount of the abrupt changes in the ridge flow.

Application Module

The application module deals with user interface. User can see several related forms according to the application. Mainly 2 applications: Bio-Vote, and Track-Me. The main form contains links for the applications. The login form allows user to enter into the particular application. Bio-Vote contains several forms according to the need. It include home pages of administrator, voter, supervisor, registration form etc. Track-Me include forms such as company registration form, adding new labourer form etc.

4.4 DATA DESIGN

Design of the file is very important factor in the system. While designing files we have to avoid data redundancy but the same time all required data should be considerable while the designing the database.

- Use the singular forms of table and column name
- Select the proper data types for each column
- Use descriptive names for tables, column and indexes.

4.4.1 TABLES

There are several tables used, some of them include Login, companyreg, voterregistration, fingerreg, emptab, criminal etc. Some of them are listed below;

- **Login: This table is used for store login details of the user.**

Field Name	Data Type	Description
Username	varchar(50)	Name of the User
Password	varchar(50)	Password of the user
Utype	varchar(50)	Type of the User

Table 4.3 Login Table

- **Companyreg:** This table is used for store the details of a company.

Field Name	Data Type	Description
Cmpname	varchar(50)	Name of the company
Lino	int	Company license no
Phone	varchar(50)	Phone number of company
Email	varchar(50)	Email of the company
Username	varchar(50)	User name of the company
Password	varchar(50)	Password of the company

Table 4.4 Companyregistration Table

- **Emptab:** This table is used for store the details of an employee.

Field Name	Data Type	Description
Cmpname	varchar(50)	Company Name
Adharno	varchar(50)	Aadhaar number of the employee
Empid	nvarchar(50)	Employee ID
Joiningdate	varchar(50)	Joining date of the employee
Id	int	Id of an employee

Table 4.6 Employee Table

- **Createsupervisor:** This table is used for store the details of the supervisor.

Field Name	Data Type	Description
Superid	int	Id of the supervisor
Boothid	int	Id of the booth
super name	varchar(50)	Name of the supervisor
Username	varchar(50)	User name of the supervisor
Password	varchar(50)	Password of the supervisor

Table 4.7 Supervisor Table

- **Fingerreg:** This table is used for storing the fingerprints.

Field Name	Data Type	Description
Id	Int	Id of the fingerprint
Cardid	nvarchar(50)	Card id of the person
Fing	varchar(2000)	Finger

Table 4.8 Fingerprint Table

- **Createstate:** This table is used for store the state details.

Field Name	Data Type	Description
Stated	Int	Id of the state
Statename	varchar(50)	Name of the State
Statecode	varchar(50)	State code of the State

Table 4.9 Create State Table

- **Createdistrict:** This table is used for store the details of the district.

Field Name	Data Type	Description
Distid	int	Id of the District
Distname	varchar(50)	Name of the District
Stated	int	Id of the State

Table 4.10 Create District Table

- **Createcandidate:** This table is used for store the details of the candidate

Field Name	Data Type	Description
Candidateid	Int	Id of the Candidate
Candidatename	varchar(50)	Name of the candidate
Cardcode	varchar(50)	Card code of the candidate
Stated	varchar(50)	Id of the state
Districtid	varchar(50)	Id of the District
Photo	varchar(MAX)	Photograph of the candidate
Symbol	varchar(MAX)	Voting Symbol of the Candidate

Table 4.11 Create Candidate Table

- **Criminal:** This table is used for store criminal history.

Field Name	Data Type	Description
Id	varchar(50)	Card ID of the User
Criminalhistory	varchar(50)	Criminal history of the user

Table 4.12 Criminal History Table

4.5 DESIGN LANGUAGE DESCRIPTION

ASP.NET

ASP.NET is a programming framework built on the common language runtime that can be used on a server to build powerful Web applications. ASP.NET offers several important advantages over previous Web development models:

Enhanced Performance

ASP.NET is compiled common language runtime code running on the server. Unlike its interpreted predecessors, ASP.NET can take advantage of early binding, just-in-time compilation, native optimization, and caching services right out of the box. This amounts to dramatically better performance before you ever write a line of code.

World-Class Tool Support

The ASP.NET framework is complemented by a rich toolbox and designer in the Visual Studio integrated development environment.

Power and Flexibility

Because ASP.NET is based on the common language runtime, the power and flexibility of that entire platform is available to Web application developers. The .NET Framework class library, Messaging, and Data Access solutions are all seamlessly accessible from the Web. ASP.NET is also language-independent.

Simplicity

ASP.NET makes it easy to perform common tasks, from simple form submission and client authentication to deployment and site configuration.

Manageability

ASP.NET employs a text-based, hierarchical configuration system, which simplifies applying settings to your server environment and Web applications. Because configuration

information is stored as plain text, new settings may be applied without the aid of local administration tools.

Scalability and Availability

ASP.NET has been designed with scalability in mind, with features specifically tailored to improve performance in clustered and multiprocessor environments. Further, processes are closely monitored and managed by the ASP.NET runtime.

Customizability and Extensibility

ASP.NET delivers a well-factored architecture that allows developers to "plug-in" their code at the appropriate level. In fact, it is possible to extend or replace any subcomponent of the ASP.NET runtime with your own custom-written component. Implementing custom authentication or state services has never been easier.

Security

With built in Windows authentication and per-application configuration, you can be assured that your applications are secure.

Language Support

The Microsoft .NET Platform currently offers built-in support for three languages: C#, Visual Basic, and JScript. The exercises and code samples in this tutorial demonstrate how to use C#, Visual Basic, and JScript to build .NET applications.

Advantages of ASP.NET

- Simplified architecture and development
- Flexible language options
- Freedom of choice in servers, tools, and components
- Integration with existing information systems
- Scalability to meet demand variations
- Flexible security model
- Improved deployment and reliability

C# (C SHARP)

C# is Microsoft's new programming language for the .NET platform. It combines some of the best features of modern programming languages such as Java, C++ or Visual Basic. C# is an object-oriented language with single inheritance but multiple interfaces per class. It supports component-based programming by properties (smart fields), events and delegates (enhanced function pointers). C# is fully interoperable with other .NET languages such as VB.NET, Eiffel.NET or Oberon.NET. C# is a simple, modern, object-oriented, and type-safe programming language that combines the high productivity of rapid application development languages with the raw power of C and C++.

Because C# supports method overloading, a class or struct can contain multiple definitions of some method, provided each has a different signature. An application can consist of multiple classes or structs. It is possible for more than one of these classes or structs to contain a method called Main. In C#, every method must be defined as a member of a class or struct. Ordinarily, the declared accessibility of a method is determined by the access modifiers specified in its declaration, and similarly the declared accessibility of a type is determined by the access modifiers specified in its declaration. For a given method of a given type to be callable, both the type and the member must be accessible. However, the application entry point is a special case. Specifically, the execution environment can access the application's entry point regardless of its declared accessibility and regardless of the declared accessibility of its enclosing type declarations. C# has a unified type system. This unified type system is called Common Type System (CTS). A unified type system implies that all types, including primitives such as integers, are subclasses of the System.Object class. For example, every type inherits a ToString() method.

By design, C# is the programming language that most directly reflects the underlying Common Language Infrastructure (CLI). Most of C#'s intrinsic types correspond to value-types implemented by the CLI framework. However, the C# language specification does not state the code generation requirements of the compiler: that is, it does not state that a C# compiler must target a Common Language Runtime (CLR), or generate Common Intermediate Language (CIL), or generate any other specific format. Theoretically, a C#

compiler could generate machine code like traditional compilers of C++ or FORTRAN; in practice, all existing C# implementations target CIL.

Internet Information Services

Internet Information Services (IIS) is the web server included in Microsoft Windows Server 2003. This will serve the requested web pages that are eventually built and displayed to the user.

DATABASE

SQL Server

SQL Server is a relational database management system (RDBMS) that uses Transact SQL to send request between a client and SQL Server. SQL Server is designed to be a client/server system. Client/Server systems are constructed so that the database can reside on a central computer, known as server, and be shared among several users. When users want to access data on the SQL Server, they run an application on their local computer, known as a client that connects over a network to the server running SQL Server. The various T-SQL statements supported by SQL Server are classified as DDL, DML and DCL statements.

DDL - Data Definition Language (create, alter, drop)

DML - Data Manipulation Language (select, insert, update, delete)

DCL - Data Control Language (grant, revoke)

TCL - Transaction Control Language (commit, rollback)

The following are the factors for which I have chosen SQL Server as the back end tool:

Advantages of using SQL Server:

- Multi-user database
- Supports RDBMS
- It's very fast.
- It's relatively easy to use.
- It's widely used

CHAPTER 5

TESTING AND RESULTS

5.1 TESTING & VALIDATION

Testing and validation includes several test and validation cases.

Case 1: Altered fingerprint identification (type1)

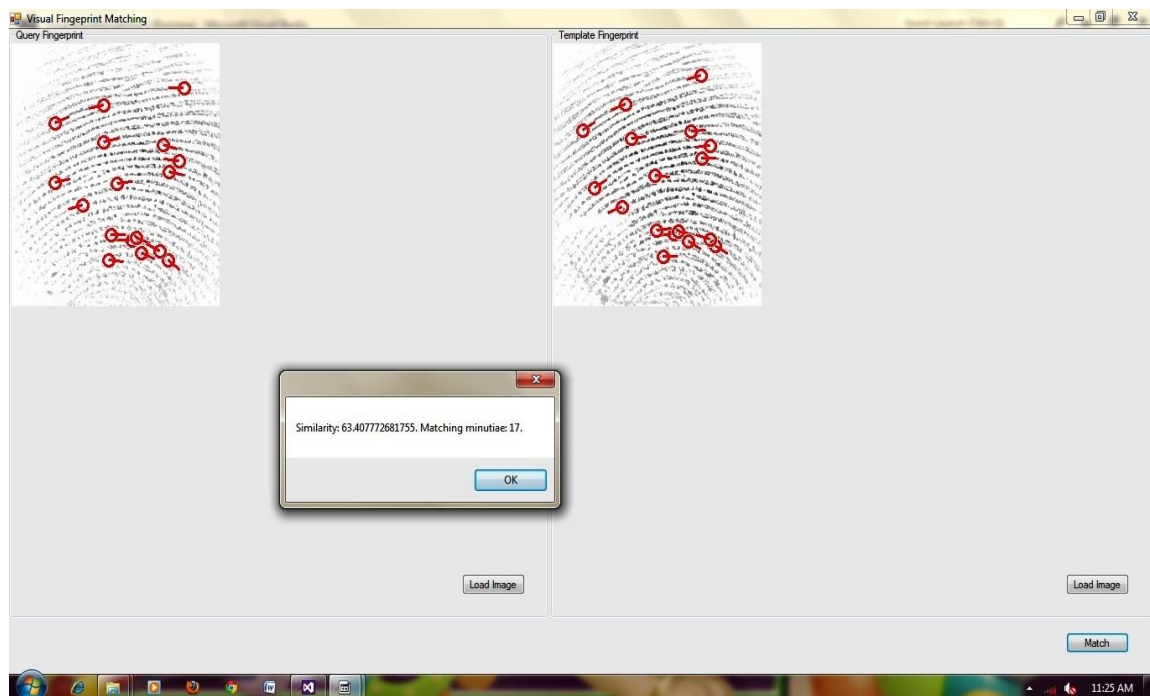


Fig:4.7 Altered fingerprint identification1

Case 2: Altered fingerprint identification (type2)

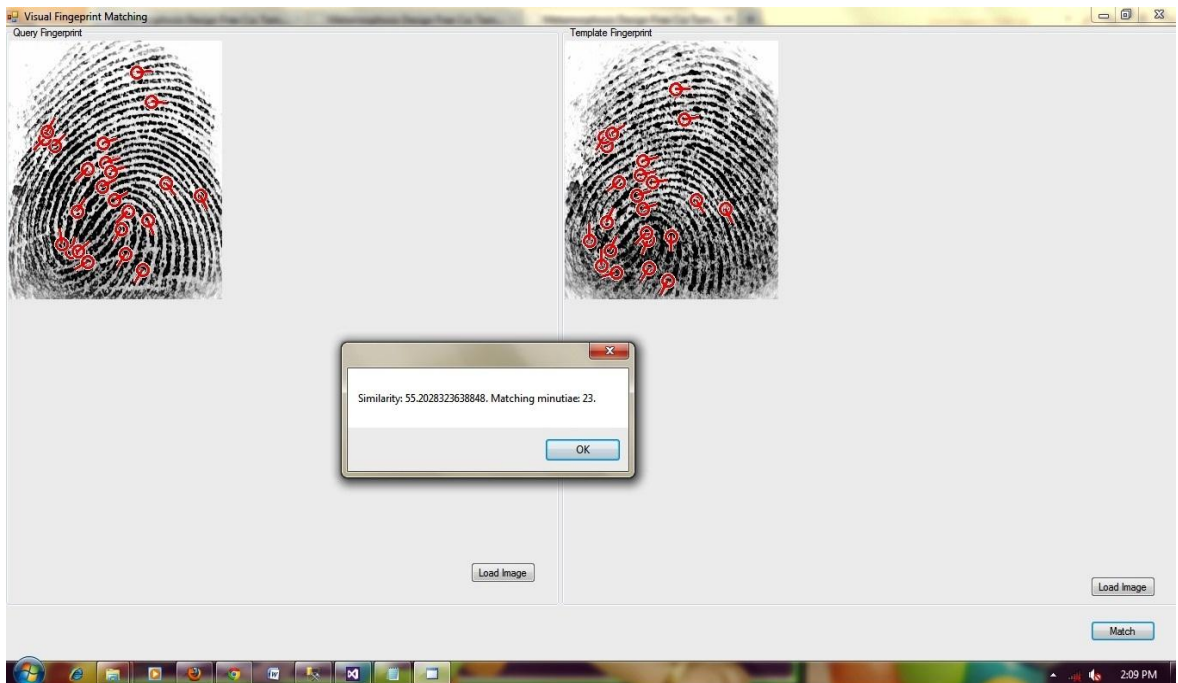


Fig:4.8 Altered fingerprint identification2

Case 3: Same (Natural) fingerprint identification(1)

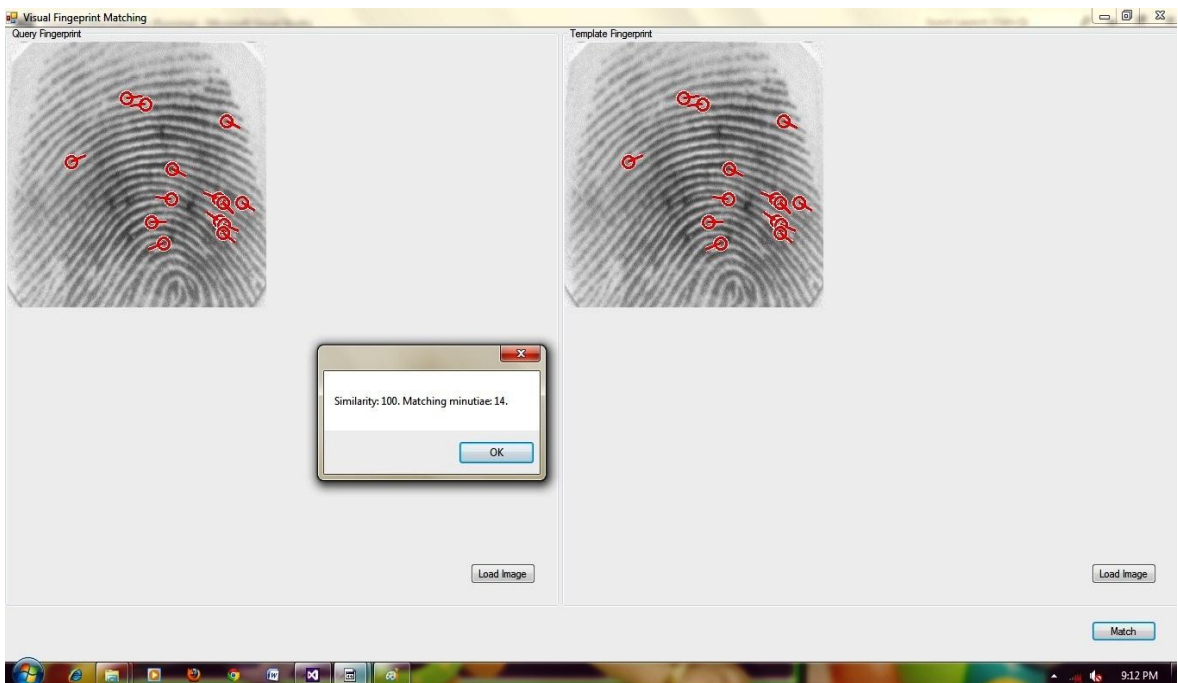


Fig:4.9 Natural fingerprint identification1

Case 4: Same (Natural) fingerprint identification(2)

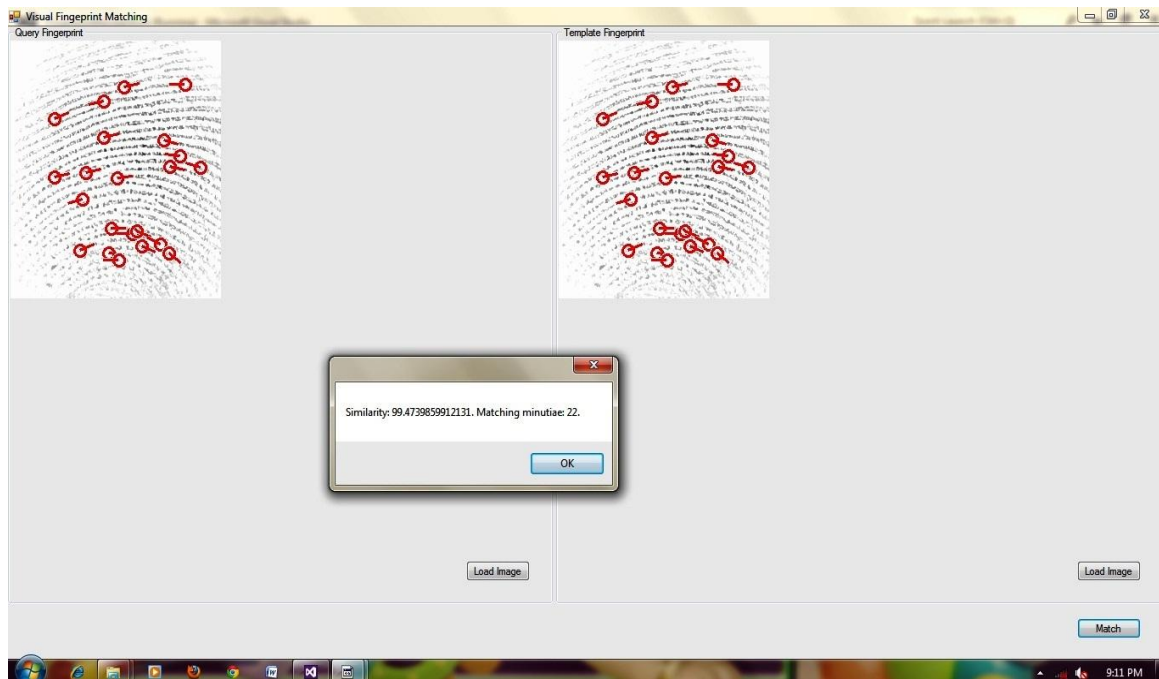


Fig:4.10 Natural fingerprint identification2

Case 5: Different fingerprints Verification

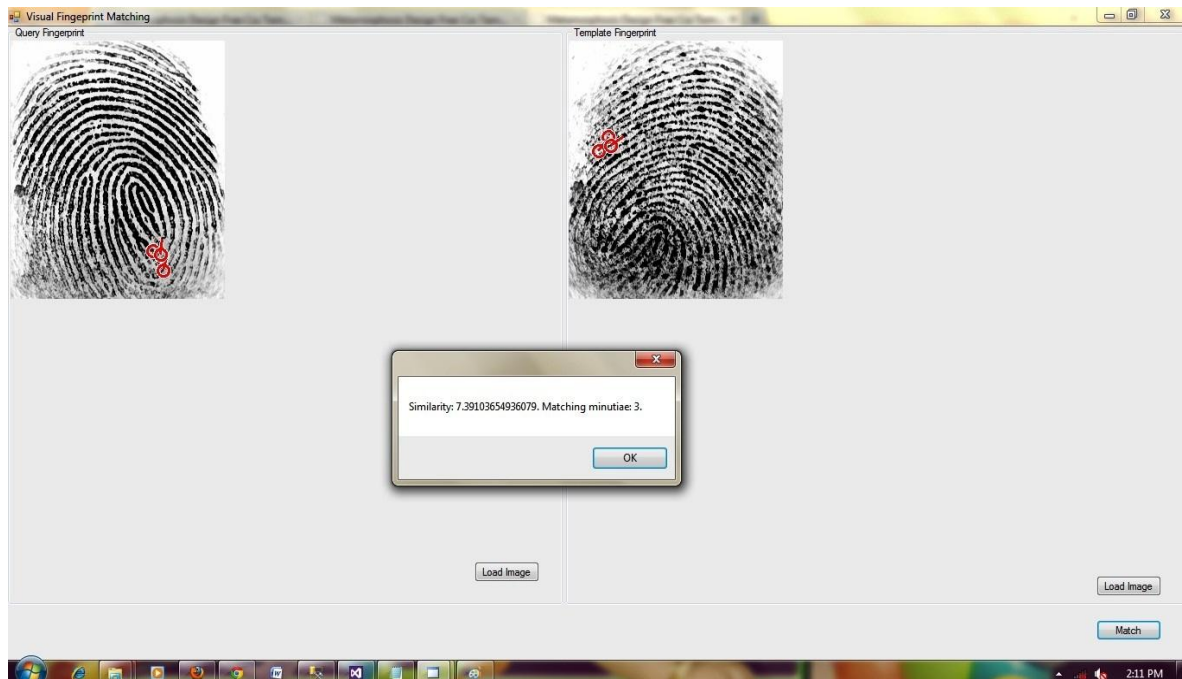


Fig:4.11 Different fingerprints verification

Case 6: Login Validation

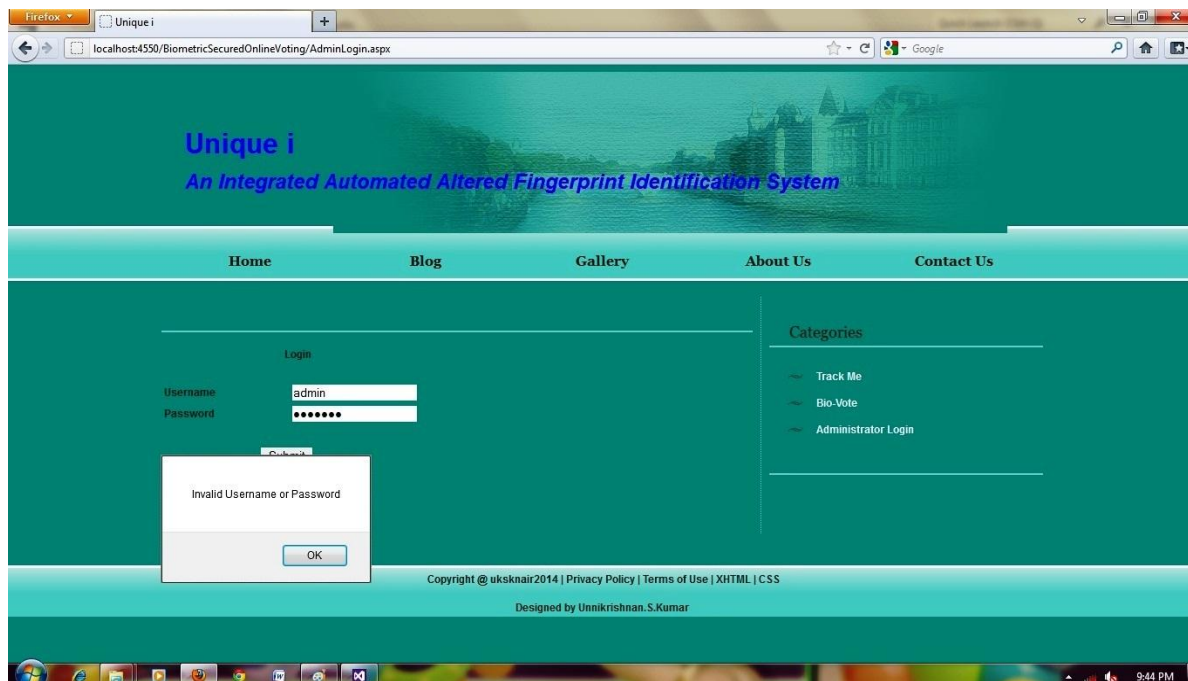


Fig:4.12 Login validation

Case 7: Create State Form Validation

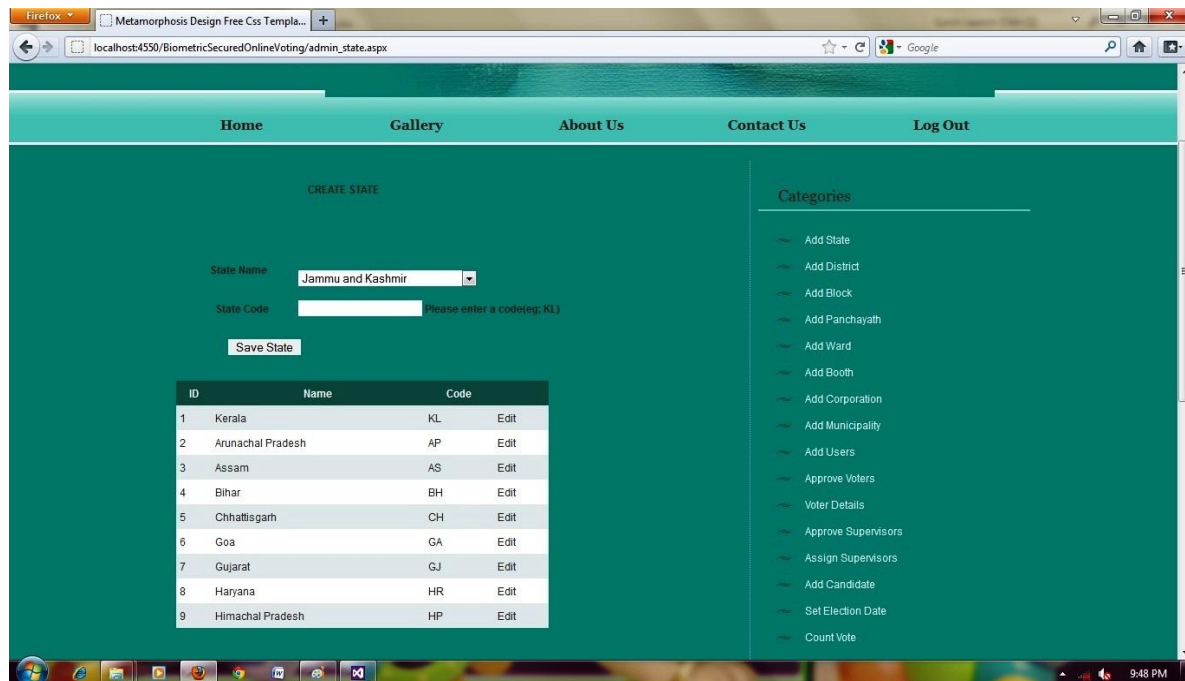


Fig:4.13 Create state form validation

Case 8: Create Candidate Form Validation

The screenshot shows a web browser window with the URL `localhost:4550/BiometricSecuredOnlineVoting/admin_candidate.aspx`. The page has a teal header with navigation links: Home, Gallery, About Us, Contact Us, and Log Out. The main content area is titled 'CREATE CANDIDATE'. It contains a form with the following fields and values:

- Candidate ID: 18
- Choose Election Spot of Candidate:
 - State: Kerala
 - District: Thrissur
 - Panchayath/Corporation/Municipality: ☐ Panchayath ☒ Corporation ☐ Municipality
 - Block: B83 Wadakkanchery
 - Panchayath: Mundathikkode
 - Ward: 12
- Candidate Details & Election Symbol:
 - Name: Hanikumar
 - Aadhaar Card No.: 34 (with a message: Please enter a valid card id (12 no.s))
 - Photo: (empty field)

On the right side, there is a 'Categories' sidebar with a list of links: Add State, Add District, Add Block, Add Panchayath, Add Ward, Add Booth, Add Corporation, Add Municipality, Add Users, Approve Voters, Voter Details, Approve Supervisors, Assign Supervisors, Add Candidate, Set Election Date, and Count Vote.

Fig:4.14 Create candidate form validation

Case 9: Company Registration Form Validation

The screenshot shows a web browser window with the URL `localhost:4550/BiometricSecuredOnlineVoting/registerCompany.aspx`. The page has a teal header with navigation links: Home, Blog, Gallery, About Us, and Contact Us. The main content area is titled 'Track Me' and contains a sub-header 'Register Your Company Now.....!!!'. It features a form with the following fields and values:

- Company Name: JK Constructions
- License No.: 79900
- Address: 885, 802, 528
- Phone No.: 999 (with a message: Please enter correct format(10 no))
- Email: jkcon (with a message: Please enter correct format(eg. abc@a.com))
- User Name: jkcon
- Password: (masked with dots)
- Re-Enter Password: (masked with dots)

Below the form is a 'Save' button. On the right side, there is a 'Categories' sidebar with links: Track Me Home, New Company registration, and Company Login. Below this is a 'Company News' section with two news items dated 05.01.2010, each with a 'read more' button.

Fig:4.15 Company registration form validation

Case 10: Employee Registration Form Validation

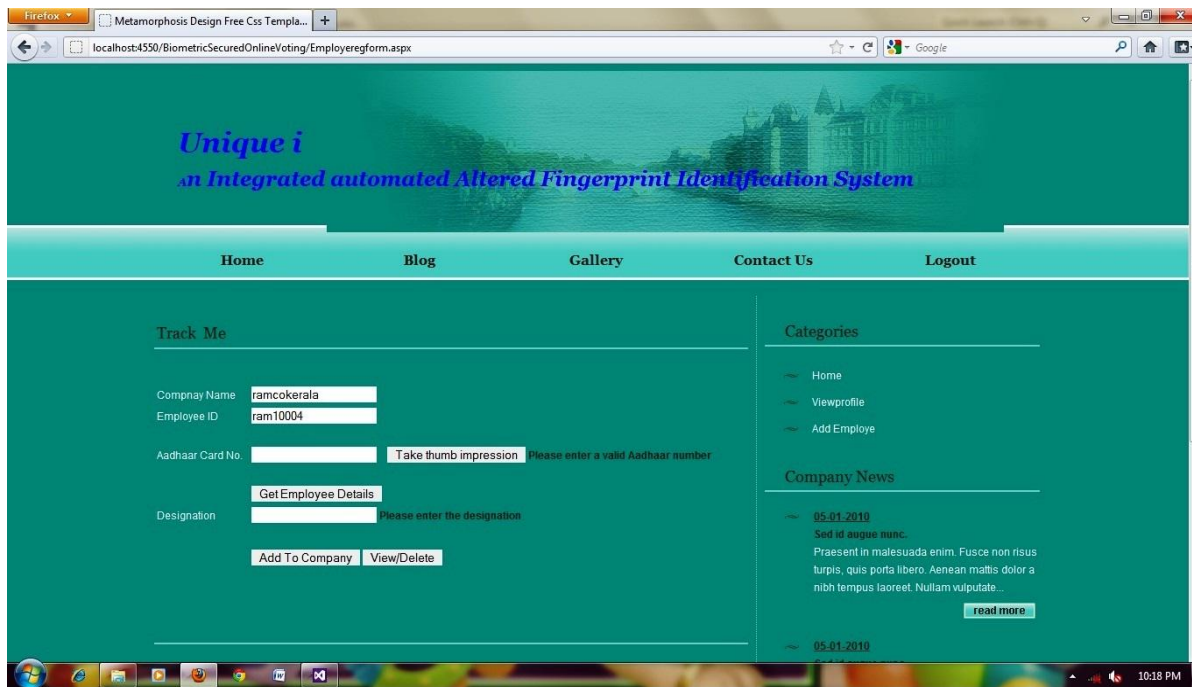


Fig:4.16 Employee registration form validation

5.2 SIMULATION & RESULTS

Unique i is a web application which integrates both fingerprint verification and identification. Live scanning of fingers are done and related outputs are shown to the user. The project contains mainly 2 phases: enrollment phase and identification phase. During the enrollment phase, the sensor scans the user's fingerprint and converts it into a digital image. The 2 applications Bio-vote and Track-Me deals with the identification phase. During the identification phase, user sees the unique i home page. This page contains the bio-vote button and track-me button. When the user clicks the bio-vote button a login form opens.

Users can login to the system using particular username and password. Users include administrator, supervisor, candidate, voter etc. The control will go to different types of pages according to the type of user. Administrator home page contains the following links such as Create/Edit/ Delete/View State, District, Block, Municipality, Corporation, Panchayath, Ward, Booth, Election date and Candidate, Approve Voters, Assign/Edit/Delete/View Supervisor,

Count Vote, View Voter Details, Change Password etc. Supervisor home page contains the following links such as View profile, Edit profile, Approve voter and store thumb impression details, Check voter, Allows voting pages to voter etc. Voter home page contains the following links such as View details, Change password, View Candidate details, Vote, Result etc. At the time of voting, the voter must touch the fingerprint sensor. When the user touches the sensor, it generates a fingerprint image. After the processing the supervisor gets the full information about the voter. If he confirms that particular voter, then the voter can vote.

When the user clicks the track-me button a login form opens. Users can login to the system using particular username and password. Users include administrator and company personnel. The control will go to different types of pages according to the type of user. Administrator home page contains the following links such as approve company profile, creation/deletion of company etc. Company profile page contains the following links such as add employee, create/delete/edit employee details etc. In the time of adding a new labourer, the fingerprint input request comes. When the labourer touches the sensor, it generates a fingerprint image. After the processing, the company personnel get the full information about the labourer. If the labourer got a criminal history or a minor (age less than 18), then the system will send a SMS to the nearest police station otherwise approve the new employee.

5.3 PERFORMANCE COMPARISON

The existing system has no identification capacity. Comparing the performance of both the existing system and the implemented system is difficult in the sense of identification. So here used a comparison method based on the time required for finding the similarity of the query print and template. Unique i is the one of the future works of the existing system. Existing system does not use a live scan method for fingerprint acquisition. The fingerprint images which are acquired using the traditional methods are scanned first. These images are stored to the database and when the verification process begins, the previous steps are repeated. For verifying the fingerprints, user has to select both the query print and template from a large database. So it takes more time as compared to the implemented system. If the size of the

fingerprint image database increases then finding the corresponding images become more and more difficult. So time factor increases.

But in the newly implemented system, there uses a live scan method for image acquisition. The image is stored in the database and in the time of identification, there is no need to select the query print and the template, they are loaded to the form automatically. So the time required to find the similarity is very less as compared to the existing system. Figure 4.17 shows a performance comparison using the histogram method. From the figure, it is clear that the implemented system, Unique i, has much more performance than the existing one.

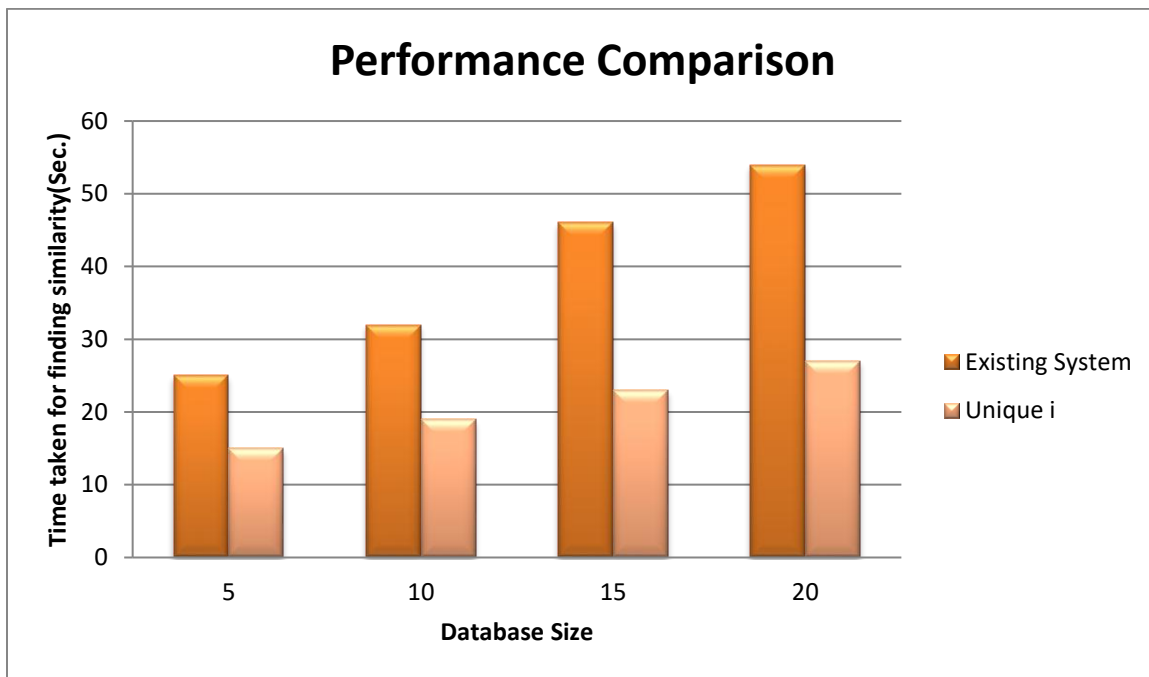


Fig:4.17 Performance comparison

CHAPTER 6

CONCLUSION

This project is a humble attempt to satisfy the user requirements & thereby overcome the problems of the existing system. It saves times and human resources space. The success of AFIS and their extensive deployment all over the world have prompted some individuals to take extreme measures to evade identification by altering their fingerprints. The problem of fingerprint alteration or obfuscation is very different from that of fingerprint spoofing, where an individual uses a fake fingerprint in order to adopt the identity of another individual. This problem is especially significant in the case of fingerprints due to the widespread deployment of AFIS in both government and civilian applications and the ease with which fingerprints can be obfuscated. The developed algorithm will automatically detect the altered fingerprints based on the characteristics of the fingerprint orientation field and minutiae distribution. The proposed algorithm based on the features extracted from the orientation field and minutiae satisfies the three essential requirements for alteration detection algorithm:

- 1) fast operational time,
- 2) high true positive rate at low false positive rate, and
- 3) ease of integration into AFIS.

In the case of Bio-Vote application, the voter's can do voting after verifying the thumb impression. Ultimately this concept tries to improve the voting and thus to improve the performance of the voters. The same thing is applicable in Track-Me application also.

In the project entitled "Unique i: An Integrated Automated Altered Fingerprint Identification System" first an attempt has been made to find the need for the system. To fulfill the needs, a detailed study has been conducted to find the various requirements of the system and it has been designed in an attractive manner, so that even a user with minimum knowledge can be able to operate the system easily. The system is well efficient and can make easy interactions with the clients. In future as time passes by, we can modify this project easily to suit the requirements of the time and further modules can be added to it.

CHAPTER 7

FUTURE SCOPE

This project can be extended in 2 levels: application level and finger level. This project only contains two types of applications. We can add more number of real time applications in several domains such as banking, border security etc. In finger level, we can use palm prints rather than a single finger which gives more security.

This study can be further extended along the following directions:

- ❖ Determine the alteration type automatically so that appropriate countermeasures can be taken.
- ❖ Reconstruct altered fingerprints. For some types of altered fingerprints where the ridge patterns are damaged locally or the ridge structure is still present on the finger but possibly at a different location, reconstruction is indeed possible.
- ❖ Use multi-biometrics to combat the growing threat of individuals evading AFIS. Federal agencies in the United States have adopted or are planning to adopt multi-biometrics in their identity management systems (the FBI's NGI and DoD's ABIS).

However, other biometric traits can also be altered successfully. It has been reported that plastic surgery can significantly degrade the performance of face recognition systems and that cataract surgery can reduce the accuracy of iris recognition systems. To effectively deal with the problem of evading identification by altering biometric traits, a systematic study of possible alteration approaches for each major biometric trait is needed.

CONFERENCES AND PUBLICATIONS

Conference Publications

[1] “Labourer Identification System.”, National Conference on Recent Advances in Computer and Electronics Engineering (NCRACEE 2014).

[2] “Unique i :An Integrated Automated Altered Fingerprint Identification System” ,National Conference on Innovations in Next Generation Computing and Communication Technology(INGCCT 2014).

Journal Publications

[1] “Unique i : An Integrated Automated Altered Fingerprint Identification System”, International Journal of Research in Computer and Communication Technology(IJRCCT), (ISSN (o)2278-5841,ISSN(p)2320-5156), Vol 3, No 3 (2014): Special Issue, March 2014.(pages 110-117)

INCLUDE THE PUBLISHED PAPER (in IEEE format) HERE



APPENDIX A

FINGERPRINTS

A fingerprint is an impression of the epidermal ridges of a human fingertip. That is it is an impression left by the friction ridges of a human finger. A fingerprint is composed of a set of lines (ridge lines), which mainly flow parallel, making a pattern (ridge pattern). Sometimes the ridge lines produce local macro-singularities, called whorl(O), loop(U) and delta.

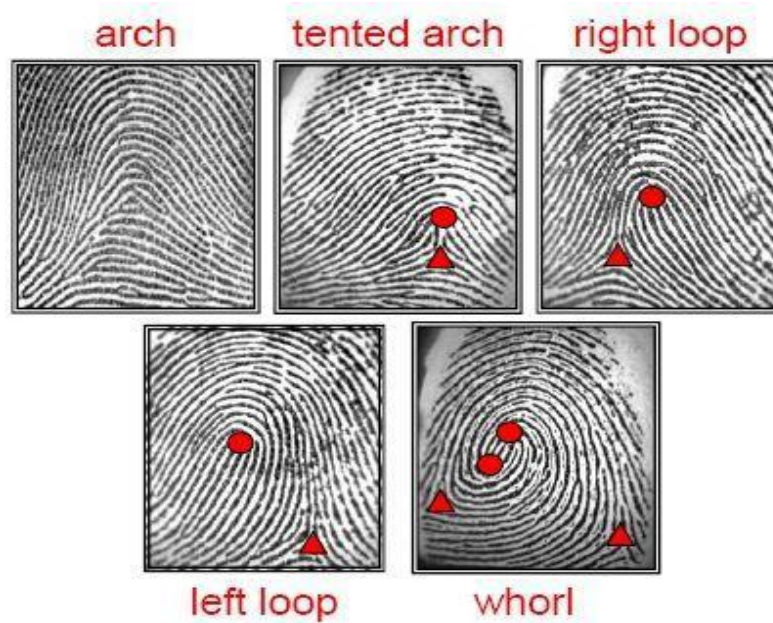


Fig. A.1: Main Fingerprint classes

A hierarchy of three levels of features, namely, Level 1 (pattern), Level 2 (minutiae points) and Level 3 (pores and ridge shape) are used for recognition purposes. Most AFISs employ Level 1 & Level 2 features. Level 1 features refer to the overall pattern shape of the unknown fingerprint - a whorl, loop or some other pattern. This level of detail cannot be used to individualize, but it can help narrow down the search. Level 2 features refers to specific friction ridge paths - overall flow of the friction ridges and major ridge path deviations (ridge characteristics called minutiae) like ridge endings, lakes, islands, bifurcations, scars, incipient ridges, and flexion creases.

MINUTIAE DISTRIBUTION

Minutiae are major features of a fingerprint, and by using this we can compare one print with another. Minutiae distribution of altered fingerprints often differs from that of natural fingerprints. Minutiae include:

- ❖ Ridge ending –the abrupt end of a ridge.
- ❖ Ridge bifurcation –a single ridge that divides into two ridges.
- ❖ Short ridge, or independent ridge –a ridge that commences, travels a short distance and then ends.
- ❖ Island –a single small ridge inside a short ridge or ridge ending that is not connected to all other ridges.
- ❖ Ridge enclosure –a single ridge that bifurcates and reunites shortly afterward to continue as a single ridge.
- ❖ Spur –a bifurcation with a short ridge branching off a longer ridge.
- ❖ Crossover or bridge –a short ridge that runs between two parallel ridges.
- ❖ Delta –a Y-shaped ridge meeting.
- ❖ Core –a U-turn in the ridge pattern.

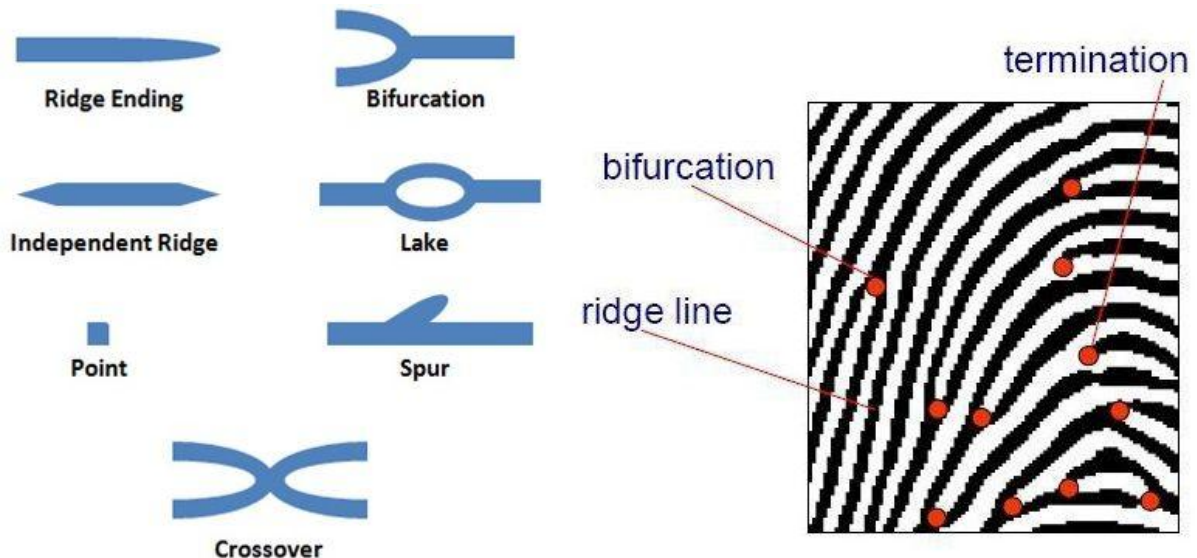


Figure A.2 : The most common minutiae (left) and a fingerprint portion

APPENDIX B

FINGERPRINT ALTERATION

Altered fingerprints are real fingers that are used to conceal one's identity in order to evade identification by a biometric system. Fingerprint alteration refers to changes made in a person's finger ridge structure by means of abrading, cutting, or performing plastic surgery on the fingertips.



Fig. B.1: Fingers burnt by acid.

The use of altered fingerprints to mask one's identity constitutes a serious "attack" against a border control biometric system since it defeats the very purpose for which the system was deployed in the first place, i.e., to identify individuals in a watch list.

Altered fingerprints are different from fake fingerprints. The use of fake fingers - made of glue, latex, or silicone - is a well-publicized method to circumvent fingerprint systems. Altered fingerprints, however, are real fingers that are used to conceal one's identity in order to evade identification by a biometric system. While fake fingers are typically used by

individuals to adopt another person's identity, altered fingers are used to mask one's own identity.



Fig. B.2: Transplanted friction ridge skin from sole.

The altered fingerprints can be classified into three categories based on the changes in ridge pattern due to alteration. This categorization will assist us in following manner: 1) Getting a better understanding of the nature of alterations that can be encountered, 2) detecting altered fingerprints by modeling well-defined subcategories, and 3) developing methods for altered fingerprint restoration.



Fig. B.3: Fingers that have been bitten.



Fig. B.4: Stitched fingers.

Obliteration

Friction ridge patterns on fingertips can be obliterated by abrading, cutting, burning, applying strong chemicals, and transplanting smooth skin. Further factors such as skin disease such as leprosy and side effects of a cancer drug can also obliterate fingerprints. Friction ridge structure is barely visible within the obliterated region. Obliterated fingerprints can evade fingerprint quality control software, depending on the area of the damage. If the affected finger area is small, the existing fingerprint quality assessment softwares may fail to detect it as an altered fingerprint, but AFIS is likely to successfully match the damaged fingerprint to the original mated fingerprint. But, if the altered area is sufficiently large, fingerprint quality control software can easily detect the damage.



Fig. B.5: Fingerprint obliteration. Examples of (a) scar and (b) mutilation.

Distortion

Friction ridge patterns on fingertips can be turned into unnatural ridge patterns by removing portions of skin from a fingertip and either grafting them back in different positions or replacing them with friction ridge skin from the palm or sole. Distorted fingerprints have unusual ridge patterns which are not found in natural fingerprints. These abnormalities include abnormal spatial distribution of singular points or abrupt changes in orientation field along the scars. Fingerprints altered by “Z” cut are of special interest since they retain their original ridge structure, enabling reconstruction of the original fingerprint before alteration. Therefore, it is imperative to upgrade current fingerprint quality control software to detect the distorted fingerprints. Once detected, the following operations may be performed to assist AFIS: 1) identify unaltered regions of the fingerprint and manually mark the features (i.e., the minutiae) in these regions and 2) reconstruct the original fingerprint as in the “Z” cut case.



Fig. B.6 : Fingerprint distortion. Examples of (a) transplantation within a finger by “Z” cut and (b) transplantation from other friction ridge skin, e.g., from the palm.

Imitation

Friction ridge patterns on fingertips can still preserve fingerprint-like pattern after an elaborate procedure of fingerprint alteration:

- 1) a portion of skin is removed and the remaining skin is pulled and stitched together,
- 2) friction ridge skin from other parts of the body is used to fill the removed part of the fingertip to reconcile with the remaining ridge structure, or

3) transplantation of the entire fingertip.

As reported in, simply swapping the skin on fingertips between the left and right hands successfully evaded AFIS. Imitated fingerprints can not only successfully pass the fingerprint quality assessment software, they can also confound human examiners. Fig.B.7 shows pre-altered and post-altered fingerprint mates. The altered fingerprint in Fig.B.7 (a) has a very smooth orientation field over the entire fingerprint area (which looks like an arch-type fingerprint) and the only evidence of possible alteration is a thin scar. The altered fingerprint in Fig.B.7 (b) was generated by an exquisite surgical procedure to have very natural ridge flow even along the surgical scars. In the case where fingerprints from different fingers are swapped, fingerprint matching without using finger position information (i.e., the left thumb is allowed to match to the right index finger) may help in determining the true identity at the expense of significantly higher matching time.



Fig. B.7: Fingerprint imitation. Left: pre-altered fingerprint, and right: its post-altered fingerprint mate. (a) Removal of a portion of skin. (b) Exquisite transplantation from other friction ridge skin.

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